

Bayesian Uncertainty Calibration for Federated Time Series Analysis

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Abstract—Deep learning models for time series analysis often require large-scale labeled datasets for training. However, acquiring such datasets is cost-intensive and challenging, particularly for individual institutions. To overcome this challenge and concern about data confidentiality among different institutions, federated learning (FL) servers as a viable solution to this dilemma by offering a decentralized learning framework. However, the datasets collected by each institution often suffer from imbalance and may not adhere to uniform protocols, leading to diverse data distributions. To address this problem, we design a global model to approximate the global data distribution of all participant clients, then transfer it to local clients as an induction in the training phase. While discrepancies between the approximate distribution and the actual distribution result in uncertainty in the predicted results. Moreover, the diverse data distributions among various clients within the FL framework, combined with the inherent lack of reliability and interpretability in deep learning models, further amplify the uncertainty of the prediction results. To address these issues, we propose an uncertainty calibration method based on Bayesian deep learning techniques, which captures uncertainty by learning a fidelity transformation to reconstruct the output of time series regression and classification tasks, utilizing deterministic pre-trained models. Extensive experiments on the regression dataset (C-MAPSS) and classification datasets (ESR, Sleep-EDF, HAR, and FD) in the Independent and Identically Distributed (IID) and non-IID settings show that our approach effectively calibrates uncertainty within the FL framework and facilitates better generalization performance in both the regression and classification tasks, achieving state-of-the-art performance.

Index Terms—Uncertainty calibration, Bayesian, time series analysis, federated learning.

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I. INTRODUCTION

TIME series analysis is vital for diverse domains like industries, healthcare, environment, and infrastructure. We can anticipate future events, detect anomalies, and classify patterns by analyzing historical time series data patterns and leveraging machine learning algorithms. These accurate detections and classification in time series analysis help optimize operational efficiency, enhance decision-making processes, prevent system failures, and facilitate proactive maintenance strategies [1], [2], [3], [4]. Given its critical importance and the advancements in sensor technology, coupled with the rapid evolution of deep learning methodologies, there has been a notable increase in research on time series analysis techniques in recent years.

Time series analysis approaches can be broadly categorized into model-based [5], [6] and data-driven methods [7], [8], [9]. Model-based approaches typically construct a mechanical model for the target system based on extensive prior knowledge for precise prediction. They typically employ mathematical theories to describe the fundamental dynamics of a system, such as differential equations or autoregressive models. These approaches are preferred when the system dynamics are well-understood. However, in practice, the complexity of systems is often unpredictable and difficult to express with mathematical equations. In such cases, these methods may struggle to accurately model the system [5], [6], [10]. In contrast, data-driven methods have gained increased attention, especially in the era of Big Data emergence. These methodologies initiate the process by extracting relevant features from raw data. Benefiting from advancements in deep learning, data-driven approaches have recently become a hot topic. However, insufficient labeled data and commercial privacy concerns in institutions hinder the widespread adoption of these approaches in real-world scenarios. Many works use few-shot deep learning techniques [11], [12], [13], [14] to address the problem of insufficient labeled data, but these methods often fail to meet the requirements of commercial privacy protection. Recently, federated learning (FL) [15] has emerged as a privacy-preserving paradigm for training models by using decentralized data from different participant clients. This facilitates model training on client data without directly access to the data. More specifically, participant clients can independently train models on their data and periodically aggregate model parameters on a central server to create a global model, ensuring data confidentiality.

The FL framework was initially proposed to solve the problem of effective learning from data distributed across multiple devices while protecting data privacy [16], [17]. In recent years, research on FL has made rapid progress, particularly in scenarios that require sensitive data processing, such as manufacturing [18], healthcare [19], [20], [21], and financial services [22], especially in time series analysis tasks with data island and privacy problems. Guo et al. [18] proposed the initial approach for incorporating the FL framework into remaining useful life (RUL) prediction. Zhang et al. [23] introduced a novel federated transfer learning method to address domain shift issues in time series fault diagnosis tasks, incorporating a federal initialization scheme, deep adversarial learning, and a prediction consistency technique to improve model accuracy while tackling low-quality distributed data issues. Lu et al. [24] proposed a class-imbalance FL method for decentralized fault diagnosis, achieving state-of-the-art performance in imbalanced classification.

However, challenges arise when dealing with non-Identically and Independently Distributed (non-IID) datasets. In these scenarios, the data distribution of local clients deviates from the global dataset, leading to convergence and model drifting issues. To address these drawbacks, Sun et al. [25] proposed FedAlign in federated machine fault diagnosis, leveraging data-free knowledge distillation to align global and local models, thereby achieving significant performance improvements over state-of-the-art methods. Li et al. [26] introduced a proximity term to enhance model convergence, particularly when facing heterogeneous local data distributions. Zhu et al. [27] proposed FedGen, an innovative FL approach that employs synthetic data generation for data augmentation and distribution alignment, overcoming limitations in local data availability and thus contributing to an improvement in FL.

The above federated time series analysis approaches are mostly designed based on FedAvg [16], which overlooks the challenge of diverse non-IID scenarios in the real world and lacks uncertainty calibration. Additionally, these methods mainly apply only to individual classification or regression tasks. Drawing inspiration from FedGen [27] while diverging from its methodology, we design a distribution transfer module, which gathers information from all participating clients to approximate the global data distribution. Subsequently, it transfers this approximate global data distribution as an inductive bias to individual local clients during the training process. By leveraging the distribution transfer module, our approach enhances generalization performance in time series regression and classification tasks.

Training deep models for time series prediction in the above FL framework would introduce uncertainty into the predicted results. (1) Discrepancies between the approximate global distribution and the actual distribution result in uncertainty in the predicted results. (2) Diverse data distributions among various clients within the FL framework would bring uncertainty in predictions. (3) The inherent lack of reliability and interpretability in deep learning models further amplifies the uncertainty in predictions. Uncertainty in machine learning can be broadly categorized into *epistemic uncertainty* and *aleatoric uncertainty*.

Epistemic uncertainty arises from the lack of knowledge about the underlying processes [28], [29], [30], while aleatoric uncertainty is associated with inherent randomness or variability in the data [31], [32], [33].

Given the presence of uncertainty in machine learning models, it is essential to employ uncertainty calibration techniques to ensure reliable and accurate predictions. These techniques can be roughly classified into ab-initio [34], [35], [36], [37] and post-hoc methods [38]. The ab-initio methods consider uncertainty during model training, starting from scratch and incorporating appropriate prior knowledge, or regularization strategies into the model structure to directly model uncertainty. At the same time, when handling large datasets and high-dimensional data, the training cost from scratch is extremely high, and these models typically do not perform as well as their deterministic counterparts [39], [40]. To settle the high cost associated with ab-initio calibration methods, a feasible approach is to post-hoc learn a transformation of the trained predictions using a calibration dataset. Recent studies [40], [41] have used the Laplace approximation to calibrate uncertainty in a post-hoc manner. However, these methods are not suitable for high-dimensional problems.

Only a few studies have researched uncertainty estimation for time series analysis. Stankeviciute et al. [42] proposed a novel extension of the inductive conformal prediction framework for time series prediction, explicitly targeting the challenge of uncertainty quantification in recurrent neural networks (RNN)-based forecasts. Kristoffer et al. [43] proposed a novel deep ensemble methodology that captures uncertainty in the interpretability of deep neural networks (DNN)-based predictions for clinical time series. This approach offers a promising avenue for enhancing the reliability of DNNs and plays a crucial role in developing robust and transparent decision support systems. These approaches are ab-initio calibration techniques, where uncertainty is explicitly modeled during training. However, ab-initio methods may suffer high computational costs, particularly when dealing with large datasets and complex model architectures. To address these challenges, BayesCap [38] proposed a post-hoc approach that provides well-calibrated uncertainty and addresses high-dimensional problems for computer vision tasks.

Different from the ab-initio calibration methods [34], [35], [36], [37], [44], we propose an innovative post-hoc calibration method to capture uncertainty of pre-trained models on both time series regression and classification tasks. This method learns a fidelity transformation that reconstructs the actual value over the output of time series regression and classification tasks, utilizing deterministic pre-trained models. By leveraging pre-trained models, our approach reduces computational overhead and maintains high performance. Additionally, our method offers flexibility in handling diverse data types and sizes, making it suitable for real-world applications with large-scale time series datasets. Overall, we propose a Bayesian uncertainty calibration method for federated time series analysis for both time series regression and classification tasks. Our contributions are as follows:

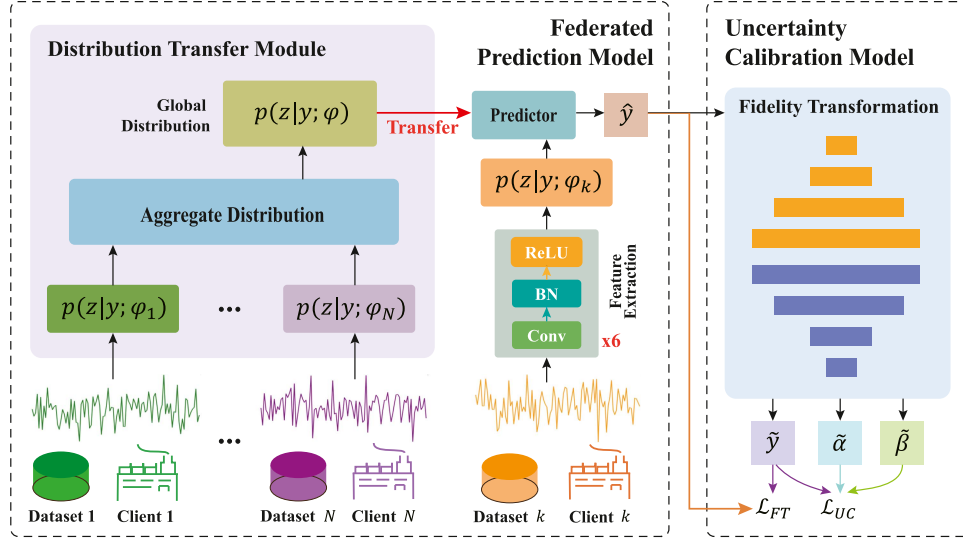


Fig. 1. The training procedure of our method. We first train a federated prediction model for time series prediction. Then, we freeze its parameters and train an uncertainty calibration model on it. During the federated prediction process, the server trains a global model to aggregate data distribution $p(\mathbf{z}|y; \varphi_i)$ from all participating clients as an approximation $p(\mathbf{z}|y; \varphi)$ to the true global data distribution. Subsequently, we transfer the aggregated global distribution to individual local clients as an inductive bias during the training process. Then, an uncertainty calibration model is proposed to obtain the uncertainty of the pre-trained federated prediction model. To achieve this, we learn a fidelity transformation to reconstruct the output $\hat{y}$ of the pre-trained model, enabling computation of calibrated uncertainty estimates. Conv, BN, and ReLU are abbreviations for convolutional layer, batch normalization layer, and rectified linear layers, respectively. $\hat{y}_i$ is the point estimate of the uncertainty calibration model. $\mathcal{L}_{FT}$ and $\mathcal{L}_{UC}$ represent the fidelity transformation loss and uncertainty calibration loss of the uncertainty calibration model, respectively.

- 1) We design a distribution transfer module for federated time series analysis, which transfers the aggregated global distribution as an inductive bias to individual local clients during the training process, with an approximate global view of client data. It successfully mitigates heterogeneity issues and facilitates superior generalization performance in both the regression and classification tasks.
- 2) We propose a post-hoc approach based on Bayesian deep learning techniques to capture uncertainty by learning a fidelity transformation to reconstruct the output of the time series regression and classification tasks, utilizing deterministic pre-trained models. This approach provides effective uncertainty calibration for the pre-trained models without hampering their performance.
- 3) The proposed method effectively mitigates data heterogeneity issues and efficiently calibrates uncertainty on the time series regression and classification tasks in the IID and non-IID settings. Furthermore, it achieves state-of-the-art performance on the time series regression and classification tasks compared to existing FL approaches.

II. PROPOSED METHOD

In this section, we first introduce the problem formulation and then describe the distribution transfer module in detail. Furthermore, we comprehensively analyze Bayesian uncertainty calibration for a pre-trained neural network. The overall architecture of our method is illustrated in Fig. 1. For the federated prediction process, the server aggregates the local label distributions from each client, forming an approximate global distribution $p(\mathbf{z}|y; \varphi)$ that covers data distribution of all local clients.

Subsequently, this approximate distribution is transferred to the clients for local training with a global data view to address data heterogeneity issue. Then, the uncertainty calibration model learns a fidelity transformation to reconstruct the output $\hat{y}$ of the pre-trained federated prediction model to obtain uncertainty in the federated prediction model.

A. Problem Formulation

Considering the following supervised federated time series prediction problem: across multiple participating clients, each client possesses its individual dataset. Let $\mathcal{D}_k = \{(\mathbf{x}_i^k, y_i^k)\}_{i=1}^{N_k}$ be the training set with pairs from the domain $\mathbf{X}$ and $\mathbf{Y}$, where N_k is the sample number of client k , $\mathbf{x}_i^k$ denotes the input signal of the i -th sample in client k from set $\mathbf{X}$, y_i^k is the corresponding label from set $\mathbf{Y}$, $\mathbf{x}_i^k \in \mathbb{R}^d$ and $y_i^k \in \mathbb{R}$. We aim to learn a global model that minimizes its loss on the participant's task T_k in a decentralized manner [16]. Therefore, the objective function, parameterized by a set of weights θ , can be defined as follows:

$$\min_{\theta} \mathbb{E}_{T_k} \left[\frac{1}{N_k} \sum_{i=1}^{N_k} \mathcal{L}_i(\theta) \right] \quad (1)$$

where $\mathcal{L}_i(\cdot)$ describes a loss function. In this work, we simultaneously consider time series regression and classification tasks, employing mean absolute error (MAE) for regression problems and negative log-likelihood (NLL) loss for multi-class classification tasks. In this manner, the global model is encouraged to learn the average optimized result of all participants' outputs. Yet effective in cases when all local datasets are generated through random and uniform allocation, adhering to the IID assumption.

However, data distributions are sometimes non-IID in practical scenarios, resulting in unsatisfactory performance.

Based on this consideration, we re-formulate the supervised federated time series prediction problem described in (1). Let $f(\mathbf{x}^k; \boldsymbol{\theta})$ represent a deep neural network parameterized by $\boldsymbol{\theta}$ that maps time series data $\mathbf{x}^k$ to label value y^k , where $f(\mathbf{x}^k; \boldsymbol{\theta})$ is trained on dataset $\mathcal{D}_k$ and is in a frozen state with parameters set to the learned optimal values $\boldsymbol{\theta}^*$. Hence, given an input $\mathbf{x}$, the model predicts a point estimate of value $\hat{y} = f(\mathbf{x}; \boldsymbol{\theta}^*)$. However, this estimation does not involve the potential model shift brought by imbalanced data distribution across clients. To address this problem, we adopt point estimation to consider the distribution of the output $p(y|\mathbf{x})$, recognizing the significance of incorporating uncertainty in time series analysis tasks [32]. Inspired by BayesCap [38], we present an approach to efficiently estimate $p(y|\mathbf{x})$ for the pre-trained model and calibrate the uncertainty associated with the predicted values without requiring model re-training. Therefore, the objective function in (1) becomes:

$$\min_{\boldsymbol{\theta}} \mathbb{E}_{T_k} [p(y^k | \mathbf{x}^k; \boldsymbol{\theta})] \quad (2)$$

B. Distribution Transfer

The presence of substantial user heterogeneity directly correlates with a heightened level of distribution divergence, consequently compromising the quality of the global model [45], [46]. Inspired by FedGen [27], we design a distribution transfer module for federated time series prediction, which transfers the aggregated global distribution as an inductive bias to individual local clients during the training process with an approximate global view of all clients data. Yet, in the task of RUL prediction, the RUL values follow a continuous distribution, posing a challenge in accurately modeling the probability distribution of these continuous labels. We discrete the RUL values by using different sampling intervals.

In this work, the global model gathers label information from all local clients to reconstruct the global distribution. To protect the data privacy of local models and avoid the high dimensionality of raw data, we learn a conditional distribution Q^* over a latent space $\mathbf{Z}$ instead of directly learning a conditional distribution over raw data:

$$Q^* = \arg \max_{\mathbf{Z}} \mathbb{E}_{y \sim p(y)} \mathbb{E}_{\mathbf{z} \sim Q(\mathbf{z}|y)} [\log p(y|\mathbf{z})] \quad (3)$$

where $p(y)$ is the prior distribution of ground truth. $Q(\mathbf{z}|y)$ is a conditional distribution of latent feature representation for an arbitrary target label y .

To obtain the prior distribution $p(y)$ in the global model, we simply calculate the number of all local client labels y through the communication between local and global models while ensuring the privacy of client data. This process is explicitly shown in (4). For the posterior probability $Q(\mathbf{z}|y)$, given an arbitrary label y , we can obtain the feature representation $\mathbf{z}$, enabling an approximate global data distribution from the ensemble of client models. Subsequently, we transfer this approximated distribution to the local clients, inducing client training by incorporating data sampled from this distribution. This posterior probability

can be approximately computed using (5).

$$\hat{p}(y) \propto \sum_{k=1}^K \mathbb{E}_{\mathbf{x} \sim \mathcal{D}_k} [\mathbf{I}(l(\mathbf{x}) = y)] \quad (4)$$

$$\log \hat{p}(y|\mathbf{z}) \propto \frac{1}{K} \sum_{k=1}^K \log p(y|\mathbf{z}; \boldsymbol{\varphi}_k) \quad (5)$$

where $l(\cdot)$ and $\mathbf{I}(\cdot)$ are a labeling function and an indicator function, respectively. K signifies the total number of participant clients. $\boldsymbol{\varphi}_k$ is the predictor parameter in the k -th client, as shown in Fig. 1.

In this manner, we can establish a distribution transfer module to alleviate the issue of user heterogeneity in time series analysis. Nevertheless, it is worth noting that the model primarily focuses on learning the average correct response for a given input, without explicitly modeling any underlying noise or data uncertainty during its estimations.

C. Bayesian Uncertainty Calibration

Bayesian deep learning techniques allow uncertainty estimation, but training them with large-scale datasets from scratch is expensive. Moreover, many high-performing deep learning models are non-Bayesian in nature and cannot provide uncertainty estimation. In this work, we introduce a post-hoc method to calibrate the uncertainty in time series regression and classification tasks, as shown in Fig. 1.

To understand our post-hoc calibration method, we first consider the time series prediction model of client k trained from scratch with uncertainty estimation. Denoting this model by $\Phi_s(\mathbf{x}^k; \boldsymbol{\phi}) : \mathbb{R}^d \rightarrow \mathbb{R}$, where $\boldsymbol{\phi}$ is a set of learnable parameters of the k -th client model. To capture the inherent uncertainty within the predictions, the model should provide estimates for the parameters of the output distribution. It is common to consider $p(y|\mathbf{x}^k)$ as a heteroscedastic Gaussian distribution [32], [33], denoted as $\{\hat{y}_i, \hat{\delta}_i\} := \Phi_s(\mathbf{x}_i^k; \boldsymbol{\phi})$, where the predicted variance serves as uncertainty in the prediction, *i.e.*, $U(\hat{y}_i) = \hat{\delta}_i^2$. Therefore, the uncertainty calibration problem can be formulated by optimizing the parameters of the Gaussian distribution.

$$\begin{aligned} \boldsymbol{\phi}^* &= \arg \max_{\boldsymbol{\phi}} \prod_{i=1}^N \frac{1}{\sqrt{2\pi\hat{\delta}_i^2}} e^{-\frac{|\hat{y}_i - y_i|^2}{2\hat{\delta}_i^2}} \\ &= \arg \min_{\boldsymbol{\phi}} \sum_{i=1}^N \frac{|\hat{y}_i - y_i|^2}{2\hat{\delta}_i^2} + \frac{\log(\hat{\delta}_i^2)}{2} \end{aligned} \quad (6)$$

The distribution of time series datasets can vary significantly depending on the specific characteristics of the data and the tasks for which they are used. It is important to note that these distributions may not always adhere to the standard Gaussian distribution. In the context of RUL prediction, heavy-tailed distributions are often observed, which can arise from premature component failures or persistent anomalies. In this case, we adopt heteroscedastic generalized Gaussian distribution [47],

[48] to model the distribution of time series regression and classification tasks within our work.

$$\phi^* = \arg \max_{\phi} \prod_{i=1}^N \frac{\hat{\beta}_i}{2\hat{\alpha}_i \Gamma(1/\hat{\beta}_i)} e^{-(|\hat{y}_i - \gamma_i|/\hat{\alpha}_i)^{\hat{\beta}_i}} \quad (7)$$

As (7) demonstrates, $\hat{y}_i$ represents the output of $\Phi_s(\mathbf{x}^k; \phi)$, while scale $\hat{\alpha}_i$, shape $\hat{\beta}_i$, and location $\hat{\gamma}_i$ are trainable parameters of the distribution, i.e., $\{\hat{y}_i, \hat{\alpha}_i, \hat{\beta}_i, \hat{\gamma}_i\} := \Phi_s(\mathbf{x}_i^k; \phi)$. $\Gamma(\cdot)$ is the Gamma function [49], i.e., $\Gamma(\tau) = \int_0^\infty t^{\tau-1} e^{-t} dt, \forall \tau > 0$. To align the outputs of $\Phi_s(\mathbf{x}^k; \phi)$ with the specific value y and simplify the interpretation, we set the location parameter $\hat{\gamma}_i = y_i$ here. Consequently, (7) is updated to (8).

$$\begin{aligned} \phi^* &= \arg \max_{\phi} \prod_{i=1}^N \frac{\hat{\beta}_i}{2\hat{\alpha}_i \Gamma(1/\hat{\beta}_i)} e^{-(|\hat{y}_i - y_i|/\hat{\alpha}_i)^{\hat{\beta}_i}} \\ &= \arg \min_{\phi} \sum_{i=1}^N \left(\frac{|\hat{y}_i - y_i|}{\hat{\alpha}_i} \right)^{\hat{\beta}_i} - \log \frac{\hat{\beta}_i}{\hat{\alpha}_i} + \log \Gamma(1/\hat{\beta}_i) \end{aligned} \quad (8)$$

$$U(\hat{y}_i) = \frac{\hat{\alpha}_i^2 \Gamma(3/\hat{\beta}_i)}{\Gamma(1/\hat{\beta}_i)}, \quad (9)$$

where $U(\hat{y}_i)$ is the uncertainty of the point estimation value $\hat{y}_i$.

So far, the optimization process estimates uncertainty with a complete retraining of the neural network, which we aim to avoid. Inspired by BayesCap [38], for a pre-trained time series prediction model $\Phi(\mathbf{x}^k; \phi^*)$ without uncertainty estimation in k -th client, we freeze its parameters first, then learn a Bayesian regression model $\Psi(\mathbf{x}^k; \psi) : \mathbb{R} \rightarrow \mathbb{R}$ on it to reconstruct the output of the pre-trained model.

$$\Psi(\hat{y}_i; \psi) = \{\tilde{y}_i, \tilde{\alpha}_i, \tilde{\beta}_i\} \quad (10)$$

$$y_i \sim \frac{\tilde{\beta}_i}{2\tilde{\alpha}_i \Gamma(1/\tilde{\beta}_i)} e^{-(|\tilde{y}_i - y_i|/\tilde{\alpha}_i)^{\tilde{\beta}_i}} \quad (11)$$

As described in (10), for every input $\mathbf{x}_i^k$, there is an output from the pre-trained model, i.e., $\hat{y}_i = \Phi(\mathbf{x}_i^k; \phi^*)$. We take it as the input of this Bayesian model.

As the point estimate $\tilde{y}_i$ of the Bayesian regression model get closer to the point estimate $\hat{y}_i$ of the pre-trained model, the uncertainty estimated by the Bayesian model also gradually gets closer to the level of uncertainty generated by the pre-trained model. So, we can extract the uncertainty $U(\tilde{y}_i)$ in a frozen pre-trained model with a Bayesian regression model.

$$\tilde{y}_i \rightarrow \hat{y}_i \Rightarrow U(\tilde{y}_i) = \tilde{\sigma}_i^2 \rightarrow \hat{\sigma}_i^2 = U(\hat{y}_i) \quad (12)$$

$$U(\tilde{y}_i) = \frac{\tilde{\alpha}_i^2 \Gamma(3/\tilde{\beta}_i)}{\Gamma(1/\tilde{\beta}_i)} \quad (13)$$

To obtain the optimal parameters ψ^* during stage training of the Bayesian model $\Psi(\hat{y}_i; \psi)$, we use the fidelity transformation loss $\mathcal{L}_{FT}$ to minimize the distance between $\tilde{y}$ and $\hat{y}$. For a regression task, the fidelity transformation loss is set to MAE; for a classification task, the fidelity transformation loss is set to NLL loss. Additionally, as we model the training data distribution as a heteroscedastic generalized Gaussian distribution, the

TABLE I
DETAILS OF THE C-MAPSS DATASET

The C-MAPSS Dataset				
Description	FD001	FD002	FD003	FD004
Training Trajectory	100	260	100	249
Test Trajectory	100	259	100	248
Operational Condition	1	6	1	6
Engine Fault Mode	1	1	2	2

TABLE II
DETAILS OF THE FOUR CLASSIFICATION TASK DATASETS

Classification Datasets				
Description	ESR	Sleep-EDF	HAR	FD
Training Sample	9200	33846	7352	43648
Test Sample	2300	8462	2947	10912
Sample Length	178	3000	128	5120
Sample Channel	1	1	9	1
Sample Class	2	5	6	3

uncertainty calibration loss is denoted as $\mathcal{L}_{UC}$.

$$\mathcal{L}_{FT} = \begin{cases} -\frac{1}{N} \sum_{i=1}^N |\tilde{y}_i - \hat{y}_i|, & Reg. \\ -\frac{1}{N} \sum_{i=1}^N \log(P(\tilde{y}_i)), & Cla. \end{cases} \quad (14)$$

$$\mathcal{L}_{UC} = \sum_{i=1}^N \left(\frac{|\tilde{y}_i - y_i|}{\tilde{\alpha}_i} \right)^{\tilde{\beta}_i} - \log \frac{\tilde{\beta}_i}{\tilde{\alpha}_i} + \log \Gamma(1/\tilde{\beta}_i) \quad (15)$$

where *Reg.* and *Cla.* represent regression and classification tasks, respectively. $\hat{y}_i$ and $\tilde{y}_i$ are the point estimate of the pre-trained federated prediction model and the Bayesian regression model, respectively.

Then, we use the two loss functions described above to optimize our model and involve a coefficient λ to balance these two loss terms. Thus, the total loss function is described as follows, which ensures our Bayesian model estimates the underlying distribution and uncertainty.

$$\mathcal{L}_{TOTAL} = \mathcal{L}_{FT} + \lambda \cdot \mathcal{L}_{UC} \quad (16)$$

III. QUANTITATIVE EXPERIMENTS

In this section, we first outline our experimental setup, including details of datasets, evaluation metrics, and experimental settings. Subsequently, we compare our method with existing FL approaches. Finally, we present ablation experiment results to demonstrate the effectiveness of our approach.

A. Datasets and Implementations

1) *Datasets*: To validate our methodology, we conduct extensive experiments on various datasets, including one dataset (details provided in Table I) for the regression task and four datasets (details provided in Table II) for the classification task.

Commercial Modular Aero-Propulsion System Simulation: The C-MAPSS dataset [50] is a collection of time series data that has been gathered from various types of aircraft engines. The data is recorded by 21 sensors installed on these engines

under different operating conditions, such as speed, temperature, vibration, and other parameters. These time series signals show the progression of aircraft engine degradation across different fault types and degradation modes. The C-MAPSS dataset is divided into four subsets: FD001, FD002, FD003, and FD004. Each subset is further divided into training and test sets, which provide operation conditions information and fault modes. Details of this dataset are provided in Table I.

Epileptic Seizure Recognition: The ESR dataset [51] contains electroencephalography (EEG) signals from 500 subjects. Each subject's signal is sampled into 4097 data points, representing 23.6 seconds of brain activity. These data points are split and randomized into 23 segments; each segment contains 178 data points, representing 1 s of brain activity. Therefore, this dataset has 11500 rows, each with 179 columns, the last column being the label, indicating five states of the subjects. However, in this paper, we simplify the problem by treating it as a binary classification task, distinguishing whether the subject has a seizure.

Sleep-EDF: The Sleep-EDF [52] is a dataset for sleep staging, which contains 197 whole-night polysomnography (PSG) recordings from 20 subjects, including EEG, electrooculography (EOG), electromyography (EMG), and event markers. The EEG, EOG, and EMG signals are sampled at 100 Hz. The dataset is generated by NASA using the European Data Format (EDF) software, and the sleep is divided into five stages: wake (W), rapid eye movement (REM), and non-REM (N1, N2, N3).

Human Activity Recognition: The HAR dataset [53] is a commonly used benchmark dataset for human activity recognition, which contains triaxial acceleration and angular velocity signals collected from smartphone accelerometers and gyroscopes. The dataset is collected from 30 volunteers performing six daily activities (walking, upstairs, downstairs, sitting, standing, lying). The dataset is divided into training and test sets; each sample contains 561 feature vectors and an activity label.

Fault Diagnosis: The FD dataset [54] is an experimental dataset for bearing fault diagnosis provided by the Institute of Design and Drive Technology of the University of Paderborn in Germany. It contains synchronous measurements of motor current and vibration signals from 32 different bearing damage states and auxiliary measurements of speed, torque, radial load, and temperature.

2) Experimental Details: In this work, we validate the effectiveness of our method using five time series datasets, including one dataset for the regression task and four datasets for the classification task. We utilize the C-MAPSS dataset for the regression task, excluding redundant columns (1, 5, 6, 10, 16, 18, 19). Given the decimal nature of temperature sensor data, it is rounded to integers to facilitate analysis. Subsequently, all samples underwent group normalization based on temperature settings to ensure consistency in the data preprocessing pipeline. Sampling intervals of 1, 2, 5, and 10 are employed for data collection. Time series data within the C-MAPSS dataset are segmented using a sliding window technique with a window length of 30 and a step size of 1. The minimum value of the maximum time step for each trajectory is 126, maintaining methodological consistency with prior research [1], [55], [56]. The

classification task datasets are divided into 80% for training and 20% for testing; all samples are normalized before training.

All experiments are repeated five times with five clients, and performance is reported as the mean value. All experiments are conducted on the PyTorch framework with an NVIDIA RTX 4090 GPU. For the federated prediction process, the local models utilize a learning rate of $1e-4$, while the global model on the server client adopts a learning rate of $1e-3$. Both models employ the Adam optimizer, and a fixed random seed is set to ensure reproducibility. The training batch size is set to 32, with 100 global communication rounds and two local model training rounds. Regarding the Bayesian calibration model, we also employ the Adam optimizer with an initial learning rate of $1e-4$. Additionally, we adopt the CosineAnnealingLR strategy. A batch size of 128 and 100 training epochs are utilized for this model.

B. Metrics

Root mean square error (RMSE) and Score are adopted to evaluate the method's performance for regression tasks [1], [55], [57]. RMSE is a standard metric for RUL prediction to measure model accuracy by calculating the mean of the squared errors between predicted and actual values, providing an assessment of model performance [1], [55]. RMSE is defined as follows. A smaller value indicates that the predicted value is closer to the actual value.

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (17)$$

where y and $\hat{y}$ denote the model prediction and actual values, and N signifies the total number of test samples.

The Score quantifies the discrepancy between the predicted and actual value and incorporates a penalty by assigning a higher score when the predicted value exceeds the actual value. This design ensures that the model's prediction remains within the machine's actual RUL, as predictions beyond the RUL could potentially lead to industrial accidents. The Score is defined as follows, with a smaller value indicating better performance.

$$\text{Score} = \begin{cases} \sum_{i=1}^N \left(e^{\frac{y_i - \hat{y}_i}{13}} - 1 \right), & \hat{y}_i < y_i \\ \sum_{i=1}^N \left(e^{\frac{\hat{y}_i - y_i}{10}} - 1 \right), & \hat{y}_i \geq y_i \end{cases} \quad (18)$$

For the classification task, we utilize four metrics to verify the performance of our approach: accuracy, precision, recall, and micro-averaged F1-score (MF1). In these four metrics, a higher value indicates better performance.

C. Data Settings

To thoroughly evaluate the performance of our method, we employ three different data distribution settings on all datasets: balanced Independent and Identically Distributed (IID), unbalanced IID, and non-IID settings. These settings reflect variations

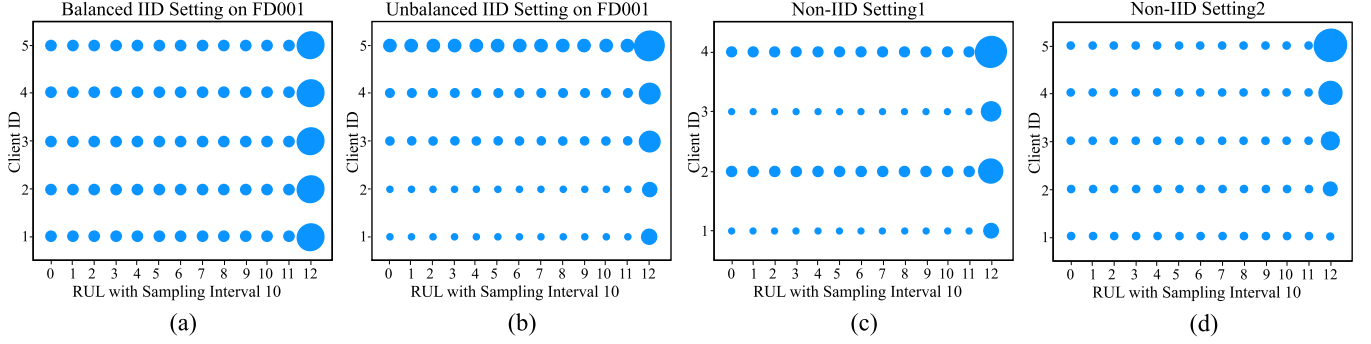


Fig. 2. Visualization for distributions of RUL with sampling interval 10 on the C-MAPSS dataset: (a) balanced IID setting, (b) unbalanced IID setting, (c) *non-IID setting1*, and (d) *non-IID setting2*. The x -axis represents RUL values with sampling interval 10, while the y -axis represents client ID. The size of the circles in the graph indicates the number of samples.

TABLE III
COMPARISON EXPERIMENTS IN THE BALANCED IID SETTING WITH DIFFERENT SAMPLING INTERVALS ON FOUR SUBSETS OF THE C-MAPSS DATASET

		FedAvg [16]		FedProx [26]		FedDistill [58]		FedGen [27]		Ours	
Dataset	Interval	RMSE↓	Score↓	RMSE↓	Score↓	RMSE↓	Score↓	RMSE↓	Score↓	RMSE↓	Score↓
FD001	1	32.87	1185.36	32.88	1178.01	33.16	1301.41	34.95	1205.28	27.16	464.99
	2	32.44	972.87	32.44	965.39	32.44	1022.62	35.41	1148.62	26.93	391.45
	5	32.96	1108.43	32.85	1088.00	32.90	1143.78	36.34	1243.53	26.67	338.60
	10	32.62	1092.45	32.66	1096.36	32.54	1117.34	36.17	1182.92	27.90	383.40
FD002	1	39.09	6789.78	39.14	6829.86	39.11	6685.20	41.38	5828.73	31.46	1926.53
	2	39.02	6864.52	39.10	6886.78	39.05	6873.52	41.38	5887.00	31.89	1906.85
	5	39.02	6986.69	38.96	6868.26	39.19	7094.89	42.38	5689.01	31.26	1657.89
	10	39.14	6885.76	39.18	6966.18	39.20	7127.66	42.58	5512.67	33.14	2253.71
FD003	1	30.72	1025.48	30.77	1048.65	30.86	1158.58	37.24	1462.18	25.96	366.54
	2	32.24	1662.69	32.25	1685.45	31.44	1595.48	38.66	1457.96	27.93	602.16
	5	30.97	1140.72	31.03	1089.67	30.72	1242.51	37.66	1322.56	26.48	405.45
	10	30.93	1112.77	30.93	1095.97	30.90	1055.58	37.86	1361.10	28.95	609.48
FD004	1	40.29	6856.53	40.38	6997.12	40.56	7154.10	43.41	5034.90	31.83	1531.17
	2	40.21	6851.17	40.32	6938.42	40.40	7446.16	43.43	4868.71	31.37	1494.00
	5	40.14	6651.83	40.13	6665.12	40.06	6726.91	43.33	4587.57	31.47	1459.41
	10	40.10	6657.98	40.16	6774.12	40.39	6943.96	43.12	4450.67	32.20	1763.31

↓ means lower values are preferred for the metric; best results in each row are in bold.

in data distribution and feature discrepancies among clients. The specific descriptions are provided below.

1) *Balanced IID Setting*: For the regression task dataset C-MAPSS, each subset's engine operational trajectories are randomly divided into five equal parts, with each client assigned one of these parts. This ensures that the data distribution among all five clients is IID. As depicted in Fig. 2(a), the data distribution among the five clients is consistent and balanced. This process is applied for the four subsets FD001, FD002, FD003, and FD004. Similarly, for the classification task datasets, we randomly divide each dataset into five equal parts based on the labels, as their label values are discrete.

2) *Unbalanced IID Setting*: Unlike the balanced IID Setting, for the C-MAPSS dataset, we randomly divided all engine operational trajectories into five parts on a fixed ratio in the unbalanced IID setting. This process is applied for the four subsets FD001, FD002, FD003, and FD004. As depicted in Fig. 2(b), the five parts have different amounts of samples. We randomly divide all samples into five parts on the same fixed

ratio for the other four classification datasets. We set the ratio as {10%, 10%, 20%, 20%, 40%}.

3) *Non-IID Settings*: For ESR, Sleep-EDF, HAR, and FD datasets, we employ the Dirichlet distribution $D(\alpha)$ to generate client data distributions. A smaller value of α indicates higher data heterogeneity. In this work, we use three values for α : 1, 5, and 10.

However, for the C-MAPSS dataset, aircraft engines are considered independent entities in real-world scenarios, and their sensor sequence data should not be distributed across multiple clients. Using the Dirichlet non-IID setting would result in sensor data from the same engine being distributed among different clients, which contradicts reality. Therefore, we use two other non-IID setting strategies. *Non-IID Setting1*: Due to differences in the number of trajectories, operational conditions, and fault modes among the subsets of the C-MAPSS dataset, we directly assign these four subsets to four clients to generate a non-IID data distribution. The distribution is illustrated in Fig. 2(c). *Non-IID Setting2*: We merge the four subsets of the C-MAPSS dataset and sort all trajectories based on the

TABLE IV
COMPARISON EXPERIMENTS IN THE UNBALANCED IID SETTING WITH DIFFERENT SAMPLING INTERVALS ON FOUR SUBSETS OF THE C-MAPSS DATASET

		FedAvg [16]		FedProx [26]		FedDistill [58]		FedGen [27]		Ours	
Dataset	Interval	RMSE↓	Score↓	RMSE↓	Score↓	RMSE↓	Score↓	RMSE↓	Score↓	RMSE↓	Score↓
FD001	1	32.43	1067.74	32.46	1089.11	32.91	1165.60	34.60	1125.83	26.44	375.02
	2	32.01	971.81	32.02	964.49	32.03	927.41	34.90	1149.40	27.02	461.98
	5	32.18	929.49	32.13	939.37	32.51	990.53	34.64	1217.00	26.34	351.91
	10	31.72	915.24	31.73	900.84	31.58	928.05	34.72	1137.93	27.29	375.40
FD002	1	38.52	6653.48	38.50	6787.32	38.34	6785.91	41.05	5876.90	30.64	1979.53
	2	40.56	6800.16	40.64	7091.60	40.76	7243.36	43.38	5803.44	32.10	1922.07
	5	38.82	6819.44	38.84	6802.44	38.76	6843.54	42.29	5688.44	31.44	1959.91
	10	38.72	6697.18	38.67	6554.65	38.67	6833.07	42.34	5271.61	33.88	2383.42
FD003	1	31.62	1111.75	31.54	1080.14	31.51	1133.75	37.26	1451.73	27.69	497.23
	2	33.14	2821.38	33.06	2521.93	34.82	3410.50	41.25	1736.78	28.55	710.35
	5	30.41	1096.13	30.43	1103.18	31.00	1258.87	36.68	1340.86	26.98	524.90
	10	32.18	2162.84	32.20	2112.07	33.02	2767.45	39.36	1405.70	30.29	792.82
FD004	1	40.51	6619.82	40.44	6618.48	40.54	6656.52	44.31	4833.44	31.39	1466.70
	2	40.30	6651.17	40.27	6632.67	40.46	6749.11	43.45	4733.13	32.15	1574.17
	5	40.81	6774.69	40.81	6800.12	41.00	6951.49	43.68	4472.07	32.47	1477.24
	10	39.89	6650.18	39.91	6716.82	39.71	6514.97	42.84	4426.19	32.52	1577.45

↓ means lower values are preferred for the metric; best results in each row are in bold.

TABLE V
COMPARISON EXPERIMENTS IN THE BALANCED AND UNBALANCED IID SETTINGS ON CLASSIFICATION DATASETS

		FedAvg [16]				FedProx [26]				FedDistill [58]				FedGen [27]				Ours			
Dataset	Setting	ACC↑	MF1↑	PRE↑	REC↑	ACC↑	MF1↑	PRE↑	REC↑	ACC↑	MF1↑	PRE↑	REC↑	ACC↑	MF1↑	PRE↑	REC↑	ACC↑	MF1↑	PRE↑	REC↑
ESR	B	81.72	53.12	83.82	54.98	81.72	53.08	82.13	54.94	81.96	54.50	83.84	55.99	<u>93.51</u>	<u>88.65</u>	<u>95.39</u>	<u>84.66</u>	94.95	91.54	95.63	88.81
	UB	81.15	52.51	79.77	54.68	81.22	52.84	79.30	54.93	81.38	53.67	81.11	55.40	<u>92.55</u>	<u>87.08</u>	<u>94.64</u>	<u>82.94</u>	93.64	89.30	94.83	86.04
Sleep EDF	B	43.17	13.56	13.89	20.98	43.13	13.50	14.73	20.96	43.13	13.35	14.40	20.89	<u>54.28</u>	<u>33.90</u>	<u>36.92</u>	<u>36.95</u>	54.73	34.51	37.55	37.59
	UB	42.99	13.51	13.13	20.96	43.07	13.69	14.18	21.15	42.87	13.17	13.36	20.74	<u>54.04</u>	<u>33.64</u>	<u>36.24</u>	<u>36.96</u>	54.61	34.38	37.04	37.78
HAR	B	20.69	11.80	19.76	21.30	20.41	12.09	24.46	20.92	20.59	11.62	17.87	21.21	<u>56.54</u>	<u>50.54</u>	56.09	<u>55.31</u>	57.69	50.63	<u>53.30</u>	56.43
	UB	20.68	11.73	20.17	21.28	20.67	11.77	20.23	21.28	20.99	12.06	20.99	21.48	<u>57.01</u>	50.46	55.79	<u>55.80</u>	57.22	<u>49.63</u>	<u>50.61</u>	55.96
FD	B	50.34	30.69	40.23	37.08	50.39	30.77	<u>41.53</u>	37.14	50.21	30.65	42.78	37.19	<u>55.01</u>	<u>36.32</u>	37.14	<u>40.14</u>	55.89	36.52	38.67	40.77
	UB	50.31	29.90	<u>44.05</u>	36.93	50.36	29.87	44.06	36.89	50.29	30.63	43.35	37.23	<u>54.76</u>	<u>35.75</u>	37.78	<u>39.97</u>	55.94	36.41	39.50	40.83

ACC, MF1, PRE, and REC represent accuracy, MACRO-F1 score, precision, and recall. B and UB represent balanced and unbalanced IID settings. ↑ means higher values are preferred for the metric, best results in each row are in bold, second best results are underlined.

TABLE VI
COMPARISON EXPERIMENTS IN THE TWO NON-IID SETTINGS WITH DIFFERENT SAMPLING INTERVALS ON THE C-MAPSS DATASET

		FedAvg [16]		FedProx [26]		FedDistill [58]		FedGen [27]		Ours	
Setting	Interval	RMSE↓	Score↓	RMSE↓	Score↓	RMSE↓	Score↓	RMSE↓	Score↓	RMSE↓	Score↓
Non-IID Setting1	1	38.84	25526.98	38.70	25619.56	35.65	20613.85	41.79	18215.62	31.04	8056.95
	2	38.79	25718.79	38.84	26442.86	35.60	21372.34	41.94	17841.15	31.13	7569.33
	5	38.89	26394.89	35.69	20555.19	38.89	26026.01	42.23	16987.71	30.99	6137.59
	10	38.95	27434.61	38.85	20822.14	35.66	21160.66	41.98	16536.94	31.82	7210.50
Non-IID Setting2	1	45.95	21233.11	45.94	20987.96	46.61	27710.30	46.42	14540.56	45.00	14018.33
	2	45.98	20795.28	46.10	21125.07	46.68	27694.73	46.53	13860.85	44.91	15828.30
	5	46.11	21087.30	46.06	21149.04	46.63	28462.53	45.79	13140.88	44.70	15358.69
	10	46.19	21555.58	46.14	21444.54	46.62	29358.60	45.39	12709.62	44.35	12483.18

↓ means lower values are preferred for the metric; best results in each row are in bold.

maximum RUL value. Finally, according to the sorting order, we evenly divide all trajectories into five parts. The distribution is illustrated in Fig. 2(d).

D. Experimental Results

In this approach, we evaluate the effectiveness of the proposed method using four representative FL algorithms: FedAvg [16],

FedProx [26], FedDistill [58], and FedGen [27]. FedAvg [16] is a standard federated averaging algorithm that independently trains local models and then averages the updates of the local models to update the global model, primarily addressing non-uniformly distributed data. FedProx [26] introduces a federated penalty term to force local models closer to the global model, thereby better maintaining the consistency of the global model, especially in the case of non-IID data. FedDistill [58] adopts the

TABLE VII
COMPARISON EXPERIMENTS IN THE NON-IID SETTINGS ON THE FOUR CLASSIFICATION DATASETS

		FedAvg [16]				FedProx [26]				FedDistill [58]				FedGen [27]				Ours			
Dataset	Setting	ACC↑	MF1↑	PRE↑	REC↑	ACC↑	MF1↑	PRE↑	REC↑	ACC↑	MF1↑	PRE↑	REC↑	ACC↑	MF1↑	PRE↑	REC↑	ACC↑	MF1↑	PRE↑	REC↑
ESR	$\alpha = 1$	88.31	69.50	70.59	71.87	88.36	69.53	72.71	71.85	79.04	65.10	67.31	71.81	91.89	74.49	<u>87.64</u>	<u>72.08</u>	92.75	84.90	95.32	83.08
	$\alpha = 5$	84.76	59.79	76.02	62.37	84.84	60.22	73.90	62.65	86.75	66.76	83.00	67.31	93.35	85.84	<u>93.34</u>	<u>82.07</u>	94.81	90.91	95.72	88.21
	$\alpha = 10$	84.89	58.97	88.21	61.26	84.96	58.80	75.69	61.03	86.42	66.40	<u>85.76</u>	66.65	93.18	77.45	<u>85.37</u>	<u>73.61</u>	94.57	88.43	93.70	85.43
Sleep EDF	$\alpha = 1$	52.63	17.49	16.31	24.70	52.64	17.51	16.40	24.71	47.65	18.01	17.70	26.41	62.48	32.85	<u>40.18</u>	<u>36.64</u>	63.06	34.45	41.38	38.03
	$\alpha = 5$	44.67	13.24	13.28	20.83	44.70	13.30	14.39	20.86	45.41	17.14	18.00	24.14	54.59	31.63	<u>36.19</u>	<u>34.88</u>	55.20	32.31	36.45	35.62
	$\alpha = 10$	43.07	13.44	13.19	20.88	43.09	13.49	13.77	20.92	43.29	13.86	15.12	21.21	54.29	33.87	<u>37.29</u>	<u>36.87</u>	54.80	34.53	37.93	37.63
HAR	$\alpha = 1$	19.84	9.46	11.80	19.89	19.83	9.42	12.49	19.88	22.64	10.61	10.23	22.19	<u>40.72</u>	31.63	<u>36.73</u>	<u>38.30</u>	47.56	35.98	37.33	44.28
	$\alpha = 5$	50.12	30.05	40.33	36.74	49.96	29.87	<u>39.09</u>	36.62	48.17	27.84	38.18	36.16	<u>54.70</u>	<u>35.96</u>	<u>37.32</u>	<u>40.05</u>	55.43	36.12	38.49	40.59
	$\alpha = 10$	19.87	10.24	16.49	20.13	19.80	10.16	16.48	20.06	22.68	13.90	17.18	23.01	51.45	43.53	49.85	<u>49.34</u>	53.73	44.15	<u>47.23</u>	51.32
FD	$\alpha = 1$	45.01	27.42	37.96	33.51	44.99	27.39	38.07	33.49	<u>56.65</u>	<u>36.17</u>	35.00	45.04	53.60	31.67	<u>38.22</u>	35.83	58.91	41.12	42.82	<u>43.24</u>
	$\alpha = 5$	50.12	30.05	40.33	36.74	49.96	29.87	<u>39.09</u>	36.62	48.17	27.84	38.18	36.16	<u>54.70</u>	<u>35.96</u>	<u>37.32</u>	<u>40.05</u>	55.43	36.12	38.49	40.59
	$\alpha = 10$	49.90	29.74	42.87	36.85	50.07	29.73	39.19	36.69	48.65	28.61	<u>39.34</u>	36.54	<u>54.71</u>	<u>36.10</u>	37.34	<u>40.06</u>	55.51	36.23	38.70	40.65

ACC, MF1, PRE, and REC represent accuracy, Macro-F1 score, precision, and recall, respectively. A smaller α indicates higher heterogeneity. $\uparrow$ means higher values are preferred for the metric, best results in each row are in bold, second best results are underlined.

concept of knowledge distillation by distilling the knowledge of the global model into local models, enhancing the model's performance on non-IID data. FedGen [27] is an innovative algorithm that employs a lightweight global generator on server client to gather knowledge from clients in a data-free manner and then distills the gathered knowledge back to the clients to enhance their local training.

For the proposed method, extensive experiments are performed on the three distinct data distributions: balanced IID, unbalanced IID, and non-IID. Four sampling intervals (1, 2, 5, and 10) are defined in the C-MAPSS dataset. Additionally, three α values (1, 5, and 10) are set in datasets ESR, Sleep-EDF, HAR, and FD to generate non-IID data.

1) *Experiments in the IID Settings:* In this paper, we conduct extensive experiments on four representative FL algorithms in two IID settings: a balanced and an unbalanced IID setting. All FL scenarios involve five clients, with each client participating in every experiment. For the C-MAPSS dataset, we discretize the continuous signal RUL values using four sampling intervals (1, 2, 5, 10) to facilitate a comprehensive analysis of the signal data.

Tables III and IV present the comparative experimental results on four subsets of the C-MAPSS in the balanced and unbalanced IID setting. Our method consistently achieves the lowest RMSE and Score across different subsets and sampling intervals compared to the other four FL algorithms, highlighting the advantages of our algorithm in the IID settings. Moreover, the results in Tables III and IV reveal that the RMSE and Score on FD001 and FD003 generally outperform other subsets. This can be attributed to these two subsets having a single operational condition and fewer fault mode categories, simplifying the data and aiding the deep learning model in capturing relevant features.

For the classification task datasets ESR, Sleep-EDF, HAR, and FD, we also compare four representative FL algorithms in both the balanced and unbalanced IID settings. Experimental results are summarized in Table V. We report accuracy,

TABLE VIII
ABLATION EXPERIMENTS IN THE BALANCED IID SETTING WITH DIFFERENT SAMPLING INTERVALS ON FOUR SUBSETS OF THE C-MAPSS DATASET

		FedGen [27] (Baseline)		FedGen [27] w/ DT (Ours)		FedGen [27] w/ DT and UC (Ours)	
Dataset	Interval	RMSE↓	Score↓	RMSE↓	Score↓	RMSE↓	Score↓
FD001	1	34.95	1205.28	28.51	593.17	27.16	464.99
	2	35.41	1148.62	27.83	503.15	26.93	391.45
	5	36.34	1243.53	27.81	421.15	26.67	338.60
	10	36.17	1182.92	28.96	499.09	27.90	383.40
FD002	1	41.38	5828.73	32.56	2595.80	31.46	1926.53
	2	41.38	5887.00	33.00	2704.60	31.89	1906.85
	5	42.38	5689.01	32.49	2054.22	31.26	1657.89
	10	42.58	5512.67	34.22	3559.14	33.14	2253.71
FD003	1	37.24	1462.18	27.76	494.68	25.96	366.54
	2	38.66	1457.96	29.88	840.52	27.93	602.16
	5	37.66	1322.56	28.34	537.36	26.48	405.45
	10	37.86	1361.10	29.92	672.91	28.95	609.48
FD004	1	43.41	5034.90	32.67	1927.48	31.83	1531.17
	2	43.43	4868.71	32.91	1894.48	31.37	1494.00
	5	43.33	4587.57	32.41	1617.42	31.47	1459.41
	10	43.12	4450.67	33.15	1856.44	32.20	1763.31

DT is our distribution transfer module. UC stands for uncertainty calibration model. $\downarrow$ means lower values are preferred. w/ is the abbreviation of with, best results in each row are in bold.

precision, recall, and MF1 score, with the best performance indicated in bold and the second best performance underlined. It demonstrates that our method achieves the best accuracy on all datasets in both the balanced and unbalanced IID settings. Moreover, our approach also generally attains the best results for the other three metrics. The results in Table V indicate that the performance of FedGen [27] is close to our method, with both significantly outperforming the other three methods.

2) *Experiments in the Non-IID Settings:* In this paper, we present the experimental results for the non-IID settings in both the regression and classification tasks. The results for the regression task are summarized in Table VI, while those for the classification task are presented in Table VII. From Table VI, it is evident that across the two types of non-IID settings with

TABLE IX

ABLATION EXPERIMENTS IN THE UNBALANCED IID SETTING WITH DIFFERENT SAMPLING INTERVALS ON FOUR SUBSETS OF THE C-MAPSS DATASET

		FedGen [27] (Baseline)		FedGen [27] w/ DT (Ours)		FedGen [27] w/ DT and UC (Ours)	
Dataset	Interval	RMSE↓	Score↓	RMSE↓	Score↓	RMSE↓	Score↓
FD001	1	34.60	1125.83	27.50	494.24	26.44	375.02
	2	34.90	1149.40	27.84	547.98	27.02	461.98
	5	34.64	1217.00	27.50	436.64	26.34	351.91
	10	34.72	1137.93	28.78	454.75	27.29	375.40
FD002	1	41.05	5876.90	31.74	2541.49	30.64	1979.53
	2	43.38	5803.44	33.81	2793.11	32.10	1922.07
	5	42.29	5688.44	32.21	2086.78	31.44	1959.91
	10	42.34	5271.61	34.73	3185.33	33.88	2383.42
FD003	1	37.26	1451.73	29.05	708.10	27.69	497.23
	2	41.25	1736.78	29.98	966.30	28.55	710.35
	5	36.68	1340.86	27.68	591.44	26.98	524.90
	10	39.36	1405.70	32.02	821.45	30.29	792.82
FD004	1	44.31	4833.44	32.72	1859.34	31.39	1466.70
	2	43.45	4733.13	33.16	1837.75	32.15	1574.17
	5	43.68	4472.07	33.52	1682.49	32.47	1477.24
	10	42.84	4426.19	33.15	1736.81	32.52	1577.45

DT is our distribution transfer module. UC stands for uncertainty calibration model. ↓ means lower values are preferred, w/ is the abbreviation of with, best results in each row are in bold.

TABLE X

ABLATION EXPERIMENTS IN THE TWO NON-IID SETTINGS WITH DIFFERENT SAMPLING INTERVALS ON THE C-MAPSS DATASET

		FedGen [27] (Baseline)		FedGen [27] w/ DT (Ours)		FedGen [27] w/ DT and UC (Ours)	
Setting	Interval	RMSE↓	Score↓	RMSE↓	Score↓	RMSE↓	Score↓
Non-IID Setting1	1	41.79	18215.62	31.85	9541.73	31.04	8056.95
	2	41.94	17841.15	31.88	8665.07	31.13	7569.33
	5	42.23	16987.71	31.56	7802.90	30.99	6137.59
	10	41.98	16536.94	32.33	8553.88	31.82	7210.50
Non-IID Setting2	1	46.42	14540.56	45.38	17071.89	45.00	14018.33
	2	46.53	13860.85	45.30	15916.81	44.91	15828.30
	5	45.79	13140.88	45.33	15536.66	44.70	15358.69
	10	45.39	12709.62	45.04	14134.52	44.35	12483.18

DT is our distribution transfer module. UC stands for uncertainty calibration model. ↓ means lower values are preferred for the metric, w/ is the abbreviation of with, best results in each row are in bold.

four different sampling intervals, our method exhibits outstanding performance in both RMSE and Score, noticeably improving the performance of RUL prediction compared to FedAvg [16], FedProx [26], FedDistill [58], and FedGen [27]. Additionally, performance in the *non-IID setting1* significantly outperforms that in the *non-IID setting2*, attributed to the stronger data heterogeneity generated by the data setup in the *non-IID setting2*. From Table VII, our method achieves the highest accuracy and MF1 score on all datasets, significantly outperforming other comparison methods. Moreover, our method generally achieves optimal performance for the other two metrics: precision and recall.

The comparison of experimental results clearly indicates that our method significantly outperforms the four representative FL methods, FedAvg [16], FedProx [26], FedDistill [58], and FedGen [27], in both the regression and classification tasks, whether in the IID or non-IID settings. This demonstrates the effectiveness and advancement of our proposed method.

E. Ablation Studies

In this subsection, we perform comprehensive ablation experiments separately for the regression and classification tasks to analyze the effectiveness of different modules within our method. All ablation experiments are conducted in both the IID and non-IID settings.

1) *Ablation Studies on the Regression Task*: In the ablation experiments for the regression task, we select the C-MAPSS dataset with four sampling intervals and conduct experiments in the balanced IID (Results in Table VIII), unbalanced IID (Results in Table IX), and non-IID settings (Results in Table X). We employ FedGen [27] as the baseline model. By integrating the distribution transfer (DT) module and the uncertainty calibration (UC). We significantly enhance our method's performance compared to the baseline model in various data settings. Additionally, we compare the model 'FedGen [27] w/ DT (Ours)' with the model 'FedGen [27] w/ DT and UC (ours)' to estimate the effects of the UC model, which utilizes the confidence level of model predictions to adjust the training direction. The UC model can substantially improve the performance of the baseline.

2) *Ablation Studies on the Classification Task*: We conduct extensive ablation experiments on the four classification datasets (ESR, Sleep-EDF, HAR, and FD) in the IID and non-IID settings. The experimental results are shown in Fig. 3, where the blue bar chart represents the baseline method, and the red bar chart represents our approach. It is evident from the figure that by incorporating the UC model, the baseline method shows a performance improvement of 1-2 percentage points across all datasets in different settings.

F. Sensitivity Analysis

We conduct the sensitivity analysis of the model's loss function to examine the impact of the hyperparameter λ on our method's performance. This analysis allows us to understand how adjustments to λ affect the contributions of the fidelity transformation loss $\mathcal{L}_{FT}$ and uncertainty calibration loss $\mathcal{L}_{UC}$ to the overall loss function. We present accuracy, precision, recall, and MF1 score metrics in the sensitivity analysis experiments. Here, we show the accuracy and MF1 score results. Fig. 4 illustrates the line chart for accuracy on the four classification datasets in different settings, and Fig. 5 shows the line chart for MF1 score. The x -axis represents different settings corresponding to various datasets, while the y -axis represents the values of the metrics.

The hyperparameter λ regulates the contribution of uncertainty calibration loss to the overall loss. Through Figs. 4 and 5, we observe that setting λ to a relatively small value has a limited impact on the method's performance. However, as λ exceeds a certain threshold, it can lead to inaccurate estimation of the parameters α and β , resulting in inaccurate uncertainty calibration. Moreover, an increase in λ reduces the proportion of fidelity transformation loss in the overall loss, causing the model's output $\hat{y}$ deviates from the label y , resulting in a decline in performance.

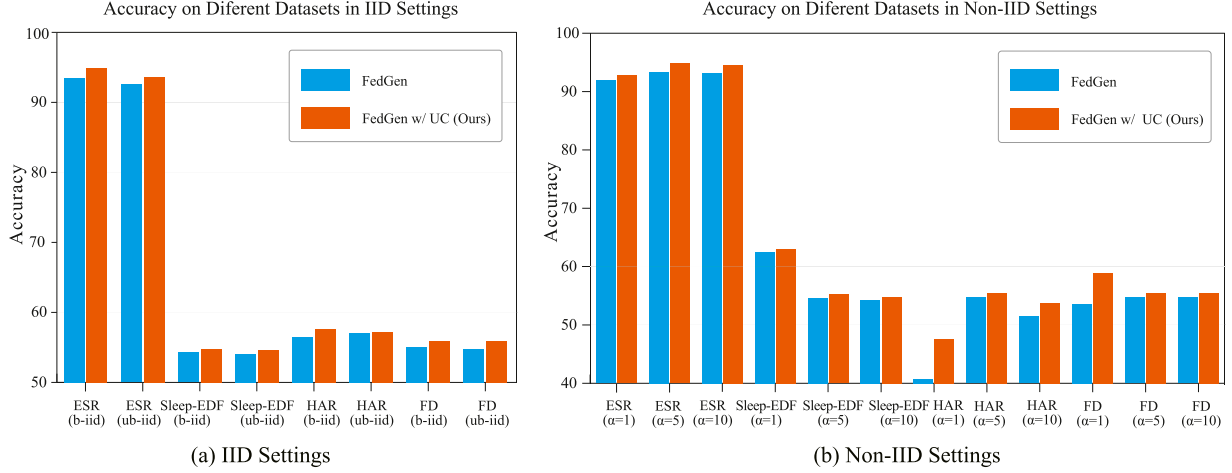


Fig. 3. Ablation study for classification tasks on the four classification datasets in the IID and non-IID settings. The x -axis represents dataset with each setting, while the y -axis represents accuracy. The blue bar chart represents the baseline method, and the red bar chart represents our approach. w/ is the abbreviation of with and UC stands for uncertainty calibration model.

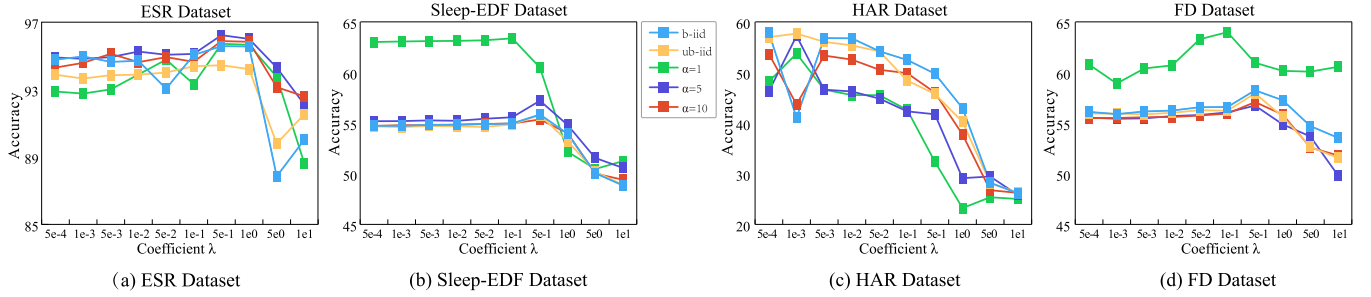


Fig. 4. Sensitivity analysis for hyperparameter λ on the four classification datasets in different settings. The x -axis represents different values of hyperparameter λ , and the y -axis represents accuracy.

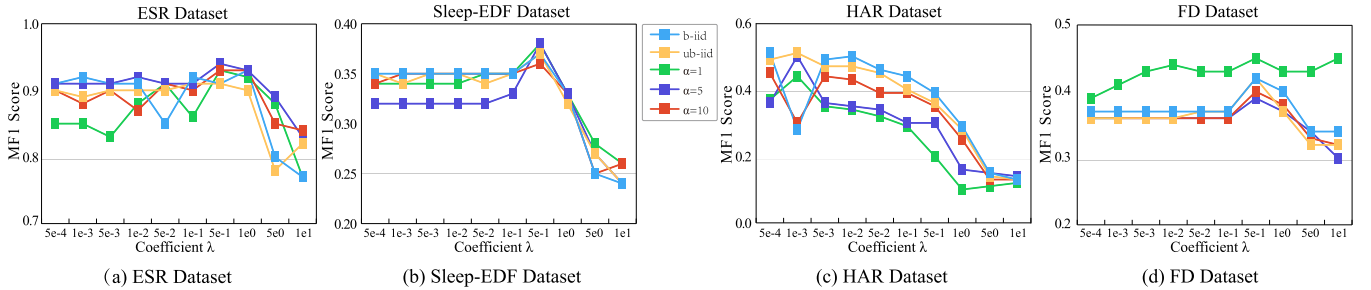


Fig. 5. Sensitivity analysis for hyperparameter λ on the four classification datasets in different settings. The x -axis represents different values of hyperparameter λ , while the y -axis represents MF1 score.

IV. CONCLUSION

This paper proposes a novel FL method with Bayesian calibration for RUL prediction and time series classification. Firstly, to address the issue of data heterogeneity in FL, we introduce an FL approach based on distribution transfer. The server aggregates the local label distributions from each client, forming a global distribution that covers the overall data distribution. Subsequently, this global distribution is transferred to the clients for

local training. Then, we present an uncertainty calibration model to capture uncertainty by learning a fidelity transformation to reconstruct the output of time series regression and classification tasks in a post-hoc manner, utilizing deterministic pre-trained models. Extensive experiments on the dataset C-MAPSS, ESR, Sleep-EDF, HAR, and FD in the IID and non-IID settings have shown the effectiveness of our method in mitigating data heterogeneous issues and effectively calibrating uncertainty for both RUL predicting and time series classification tasks. Our method

achieves state-of-the-art performance compared to existing FL approaches.

Although the proposed approach demonstrates advantages in RUL prediction and time series classification tasks, there are some shortcomings. First, we use discrete distribution to approximate the continuous distribution of RUL values in the distribution transfer module, which may not accurately capture the true distribution of RUL. Second, Although this approach can effectively capture aleatoric uncertainty, it can not capture epistemic uncertainty. In our future work, we will focus on addressing these two limitations by exploring methods to accurately capture the true distribution of continuous RUL and incorporating techniques to capture epistemic uncertainty.

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